

LAB GUIDELINES

Lab 13 — Data Migration and Transfer

Maps to CompTIA Server+ objective **2.3** — Data migration and transfer: infiltration, exfiltration, disparate OS data transfer, Robocopy, file transfer, fast copy, SCP.

Run all commands on <https://killercoda.com/playgrounds/scenario/ubuntu>

Required software (free, apt): rsync, openssh-client, openssh-server, lftp, pv

```
apt update && apt install -y rsync openssh-client openssh-server lftp pv
systemctl start ssh
```

Step 1 — Set up source and destination trees

```
mkdir -p /srv/src /srv/dst
for i in $(seq 1 50); do dd if=/dev/urandom of=/srv/src/file$i.bin bs=1K count=$((RANDOM%200+10)) 2
du -sh /srv/src
```

Step 2 — `rsync` — the workhorse for server-to-server transfer

```
rsync -avh /srv/src/ /srv/dst/
rsync -avh --dry-run --delete /srv/src/ /srv/dst/ # show what *would* sync
rsync -avh --partial --progress /srv/src/ /srv/dst/
```

-a = archive (preserves perms, times, symlinks). --delete makes dst an exact mirror. --partial resumes interrupted transfers.

Over the network:

```
rsync -avzh -e ssh /srv/src/ user@remote:/srv/dst/
```

Step 3 — SCP (Secure Copy Protocol)

```
scp -r /srv/src/ root@127.0.0.1:/srv/dst-scp/ # accept the host key
scp -p file1.bin root@127.0.0.1:/tmp/ # preserve timestamps
```

SCP is simple but **no resume**, **no delta**. Use it for one-off small copies; use rsync for bulk and ongoing sync.

Step 4 — SFTP (interactive file transfer over SSH)

```
sftp root@127.0.0.1 <<'EOF'
ls /srv
lcd /tmp
put /etc/hostname
quit
EOF
```

Step 5 — Disparate-OS transfer (Linux ↔ Windows)

Server+ explicitly mentions **Robocopy** — that is the Windows equivalent of rsync. The cross-OS transfer choices are:

Direction	Recommended tool
Linux → Linux	`rsync` (over SSH)
Windows → Windows	**Robocopy** (`robocopy src dst /MIR /Z /R:3`)
Windows → Linux	SMB mount + `rsync`, or WinSCP
Linux → Windows	Samba share + Robocopy, or `rsync` to Cygwin/WSL

```
# Linux side: mount the Windows share, then rsync into it
# mount -t cifs //win/share /mnt/win -o username=admin,password=...
```

```
# rsync -avh /srv/src/ /mnt/win/dst/
```

Step 6 — Fast copy of very large files

```
# Show throughput while copying:
pv /srv/src/file1.bin > /srv/dst/file1.bin

# Direct I/O (skip the page cache) for huge migrations:
dd if=/srv/src/file1.bin of=/srv/dst/file1.bin bs=1M oflag=direct status=progress

# Multi-threaded local copy with rsync + xargs:
ls /srv/src | xargs -n1 -P8 -I{} cp /srv/src/{} /srv/dst/
```

Step 7 — FTP / FTPS / SFTP via `lftp` (legacy compatibility)

```
# lftp -e "set ftp:ssl-allow yes; mirror -R /srv/src /backup; quit" ftpuser@ftp.example.com
echo "lftp script syntax shown – no public FTP server reachable from Killercode by default"
```

Step 8 — Infiltration vs. exfiltration awareness

Server+ 2.3 lists **infiltration** and **exfiltration** under data transfer. The same `rsync` / `scp` commands that legitimately migrate data also exfiltrate it. Controls:

1. **Egress firewall**: outbound SSH/SFTP only to whitelisted destinations (Lab 9).
2. **DLP** monitoring on file-share access (objective 3.4).
3. **Auditd watch** on sensitive directories (Lab 22).
4. **Bandwidth caps** on bulk-copy users (`rsync --bwlimit=1024`).

```
rsync -avh --bwlimit=1024 /srv/src/ user@remote:/srv/dst/
```

Free online tools

- **rsync manual** — <https://rsync.samba.org/>
- **OpenSSH manual** — <https://www.openssh.com/manual.html>
- **Microsoft Robocopy reference** — <https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/robocopy>
- **WinSCP (free, Windows)** — <https://winscp.net/>

What you learned

- `rsync` archive + delta + delete + bandwidth-limit options.
- When to use SCP (simple) vs. SFTP (interactive) vs. `rsync` (bulk).
- Cross-OS choices including the Windows **Robocopy** equivalent.
- Fast-copy techniques for very large files (`pv`, `dd oflag=direct`, parallel `xargs`).
- How the same transfer tools enable both legitimate migration **and** exfiltration, and the controls that distinguish them.